IRIS FLOWER

CLASSIFICATION

USING MACHINE LEARNING

DATA SCIENCE INTERNSHIP PROJECT

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INTRODUCTION

Iris Flower Classification is a machine learning project used to identify different types of Iris flowers.

The project uses flower measurements such as sepal length, sepal width, petal length, and petal width to predict the flower species.

Machine learning algorithms are used to classify the flowers automatically.

PROBLEM STATEMENT

Identifying different types of Iris flowers manually can be difficult. The aim of this project is to develop a machine learning model that can classify Iris flowers into different species based on their measurements.

OBJECTIVES OF THE PROJECT

- To understand the basics of Data Science.
- To learn how machine learning works.
- To explore the Iris flower dataset.
- To train machine learning models.
- To compare the accuracy of different algorithms.
- To predict Iris flower species accurately.

DATASET DESCRIPTION

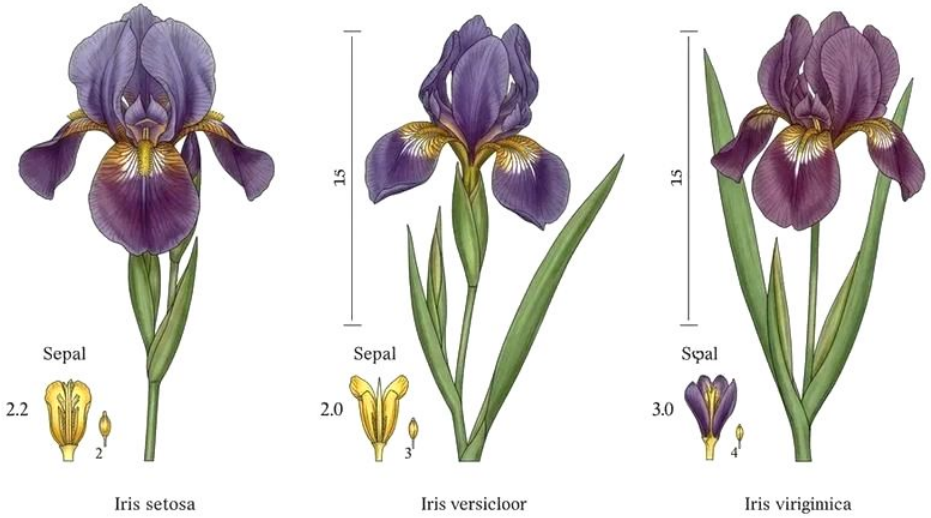
The Iris dataset contains 150 flower records.

It has four input features:

1. Sepal Length
2. Sepal Width
3. Petal Length
4. Petal Width

It contains three flower species:

- Iris Setosa
- Iris Versicolor
- Iris Virginica



The dataset is used to train and test the machine learning models.

TOOLS AND TECHNOLOGIES USED

- **Python:** Programming language used for the project.
- **Google Colab:** Online platform used to write and execute Python code.
- **Pandas:** Used to handle and organize dataset information.
- **Scikit-learn:** Used to build and test machine learning models.
- **Matplotlib:** Used for data visualization.

MACHINE LEARNING ALGORITHMS

Two machine learning algorithms were used in this project.

1. Decision Tree

A Decision Tree is an algorithm that makes predictions using a tree-like structure of decisions.

2. K-Nearest Neighbors (KNN)

KNN is an algorithm that classifies data based on the nearest data points.

Both algorithms were trained and tested using the Iris dataset.

IMPLEMENTATION STEPS

The project was completed using the following steps:

1. Imported the required Python libraries.
2. Loaded the Iris dataset.
3. Explored the dataset.
4. Divided the dataset into training and testing data.
5. Trained the Decision Tree model.
6. Trained the KNN model.
7. Tested both models.
8. Compared their accuracy
9. Predicted the Iris flower species using both models.

RESULTS AND ACCURACY

The two machine learning models produced the following results:

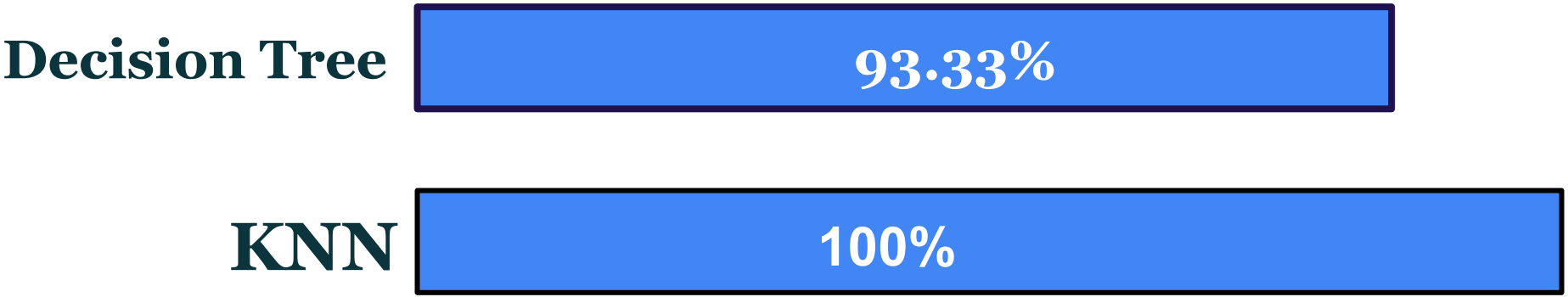
Decision Tree Accuracy: 93.33%

KNN Accuracy: 100%

The **KNN** model achieved higher accuracy than the Decision Tree model in this test.

This shows that both algorithms can be used for Iris flower classification.

COMPARISON OF ALGORITHMS



Observation:

KNN performed better than Decision Tree on the given test dataset.

conclusion

The Iris Flower Classification project helped me understand the basic concepts of Data Science and Machine Learning.

I successfully implemented two machine learning algorithms: Decision Tree and KNN.

The KNN model achieved 100% accuracy, while the Decision Tree model achieved 93.33%.

This project improved my understanding of Python, datasets, and machine learning algorithms.

FUTURE SCOPE

- Use more machine learning algorithms.
- Test the models with different datasets.
- Improve model performance.
- Develop a simple user interface for flower prediction.
- Apply machine learning to other classification problems.
- Develop a mobile application for flower identification.
- Use image recognition to identify flower species.
- Increase the size of the dataset for better predictions.
- Deploy the model on a cloud platform.
- Make the system more user-friendly and accessible.
- Compare different models to find the best algorithm.



THANK YOU

Presented as part of my Data Science Internship Project.
Project: Iris Flower Classification

